Data Science and Artificial Intelligence

Probability

Lecture Number- 03

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Topics

to be covered

1. Introduction to Probability

QUESTION

Time : 3 min

Q. 4 Apples and 3 Oranges are randomly placed in a line. Find the chance that the two extreme fruits are both oranges.

QUESTION

Time : 3 min

Q. Two natural numbers are randomly selected from the set of first 20 natural numbers. Find the probability that:

(i) their sum is odd

(ii) sum is even

(iii) selected pair is twin prime

QUESTION

Time : 3 min

Q. Bag $\left\langle \begin{matrix} 5G \text{ balls} \\ 4W \text{ balls} \end{matrix} \right.$ $\xrightarrow[\text{drawn}]{3\text{balls are}}$ find the odds against these being all green.

QUESTION

Time : 3 min

Q. 3 vertices of a regular hexagon are joined randomly. Probability that an equilateral A is formed.

QUESTION

Time : 3 min

Q. What is the chance that the fourth power of an integer chosen randomly ends in the digit six.

QUESTION

Time : 3 min

Q. 2 numbers are randomly selected from the set of first six natural numbers. Find the probability that the selected pair is coprime.

QUESTION

Time : 3 min

Q. Three tickets are drawn randomly from a pack of 51 tickets numbered from 1 to 51. Find the probability that :

(i) numbers are in AP.

(ii) their sum is odd.

QUESTION

Time : 3 min

Q. There are 10 Keys, 3 keys are randomly selected to open a lock. Find the chance that you have choosen the correct key.

QUESTION

Time : 3 min

Q. A pack of well shuffled 52 cards are randomly distributed equally among four brothers. Find the probability that all the queens are held by the youngest brother?

QUESTION

Time : 3 min

Q. Let x and y are 2 randomly selected integers. Find the probability that :

(i) $(x + y)$ is divisible by '7'.

H.W

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(ii) $(x^2 + y^2)$ is divisible by '3'.

(iii) $(x^2 + y^2)$ is divisible by 5.

QUESTION

Time : 3 min

Q. A pair of dice is thrown and shows different faces.
Find the probability
(i) sum of two faces is prime
(ii) one die shows '1'.

✓ H.W

QUESTION

Time : 3 min

Q. Let P_i (i= 1, 2, ..., 6) denotes the probability of getting the face 'i', when a biased die is thrown where $P_i \propto i$.

Find the probability of getting:

(i) Composite faces.

(ii) When 2 such dice are thrown then find the probability of getting a doublet.

H.W

QUESTION

Time : 3 min

Q. When 3 normal dice are thrown, find the probability that

(a) sum on the faces is less than 11.

(b) sum on the faces is 6.

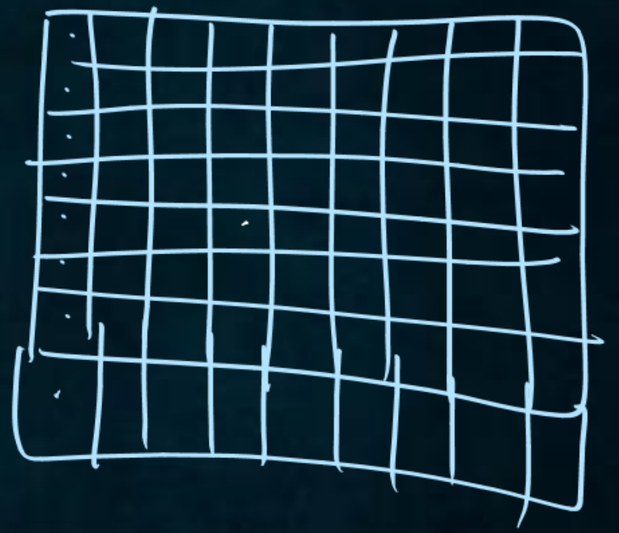
(c) sum on the faces is 9.

H.W

QUESTION

Time : 3 min

Q. 2 squares are chosen on a chessboard. Find the probability that they have a side in common.



" 2 Squares Are chosen"

Total square = 64

Total No. of outcomes $= {}^{64}C_2$

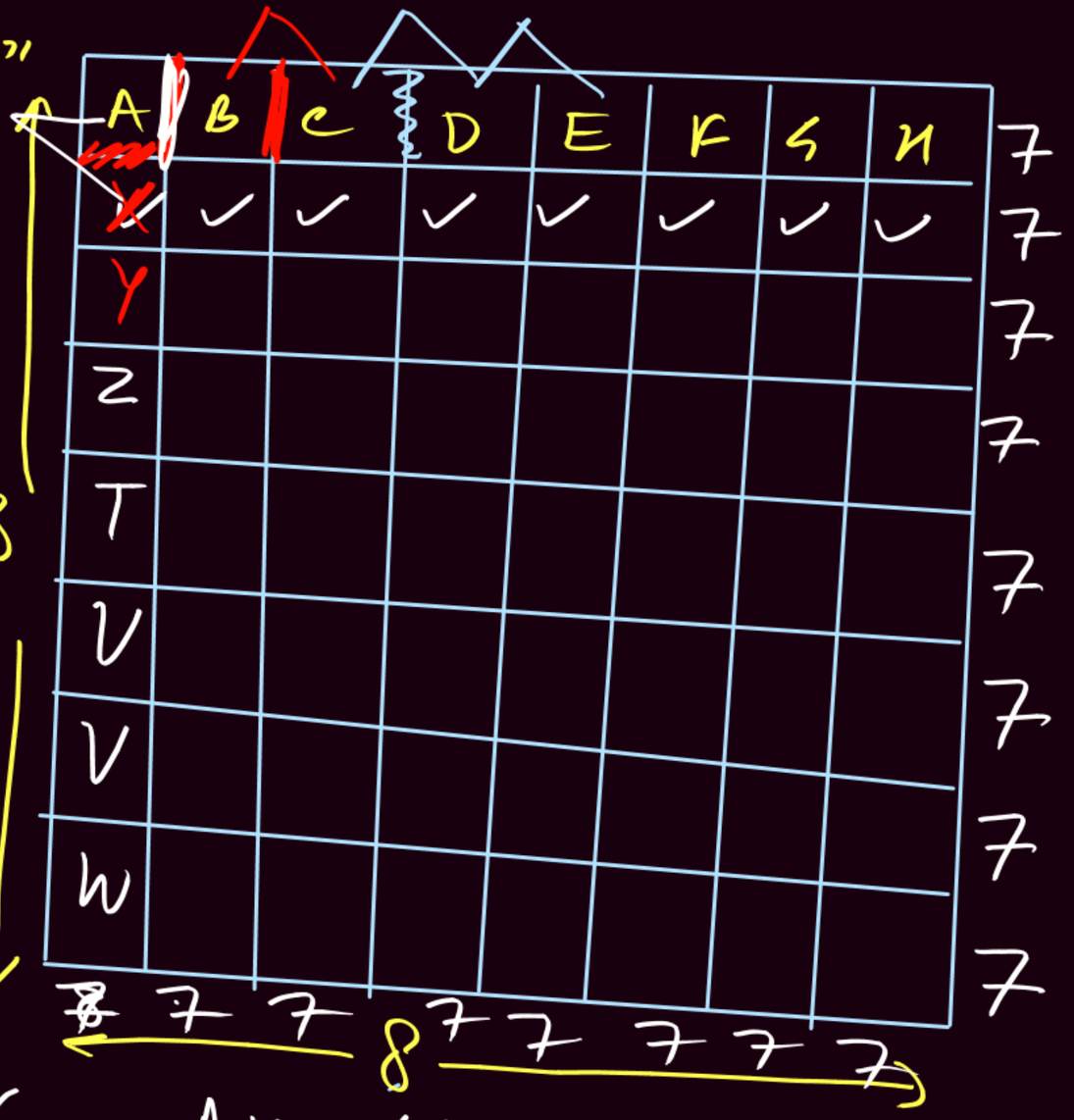
✓

AB
BC
CD
DE
EF
FG
GH

7 Ways.

Horizontal $= 8 \times 7 = 56$

Vertical $= 8 \times 7 = 56$



AX, XY, YZ, ZT, TV, VV, VW

7 Ways.

Favourable No. of outcomes = 56 + 56

= 112

$$P(E) = \frac{112}{{}^{64}C_2}$$ Ans

$${}^{n}C_r = \frac{\lfloor n}{\lfloor n-r \lfloor r}$$

$$= \frac{1}{18}$$

Ans

QUESTION

Time : 3 min

Q. Four fair dice D1, D2, D3, D4, each having six faces numbered 1, 2, 3, 4, 5, 6 are rolled simultaneously. The probability that D4 shows a number appearing on one of D1, D2 and D3 is_____.

A 91/216

B 108/216

C 125/216

D 127/216

QUESTION

Time : 3 min

Q. Seven white balls and three black balls are randomly placed in a row. The probability that no two black balls are placed adjacently_____.

A 1/2

B 7/15

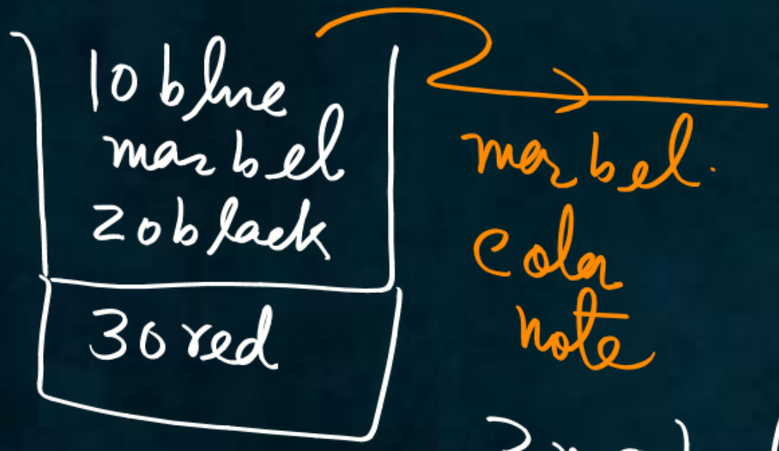
C 2/15

D 2/3

QUESTION

Time : 3 min

Q. A bag contains 10 blue marbles, 20 black marbles and 30 red marbles. A marble is drawn from the bag, its colour is recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour.



with replacement

$W = W_1 \times W_2 \times W_3$

$= m \times n \times p$

W	W_1	W_2	W_3	
				Ist choice
✓				IInd choice
✓				3rd choice
✓				4th choice
✓				5th choice
✓				6th choice

$$= \frac{10}{60} \times \frac{20}{60} \times \frac{30}{60}$$

$$= \frac{30}{60} \times \frac{10}{60} \times \frac{20}{60}$$

THREE Diff. Items

If n Diff. Items Taken all at a time

Arrangement $= n!$

A B C

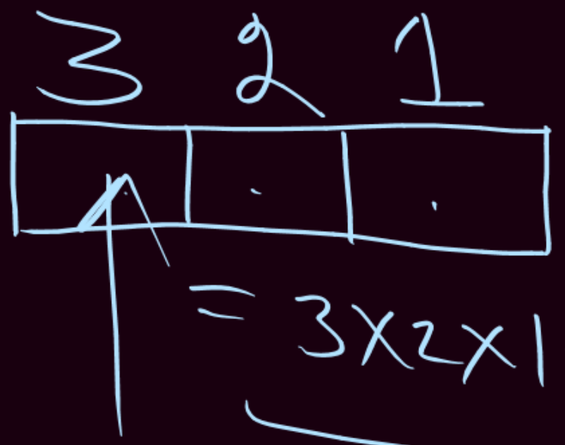
C B A

C A B

A C B

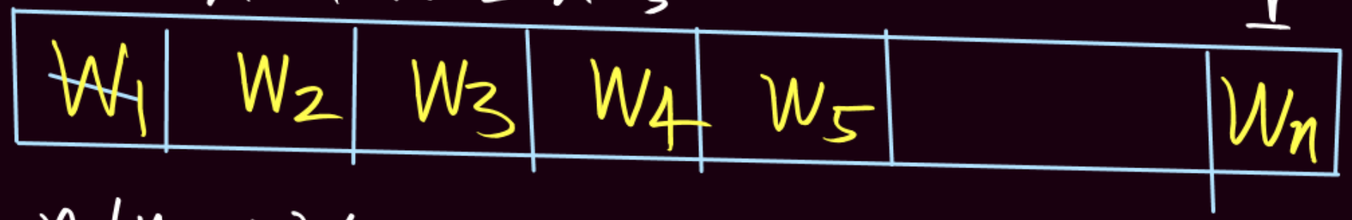
B C A

B A C



$= 3 \times 2 \times 1 = 6$

A, B, C - - -



$= n(n-1)(n-2)(n-3) \cdots 1.$

If working Together.

$W = W_1 \times W_2 \times W_3 \times \cdots W_n$

$= n(n-1) \cdots 1$

$= n!$

$= \frac{30}{60} \times \frac{20}{60} \times \frac{10}{60} \times 3! = \frac{1}{6}$ Ans

Important Note:

n Items Taken all at a time $= n!$

A B C Taken 2 at a time

AB BC CA
BA CB AC
= 6 Ans

box method

3	2

$= 3 \times 2 = 6$

A B C

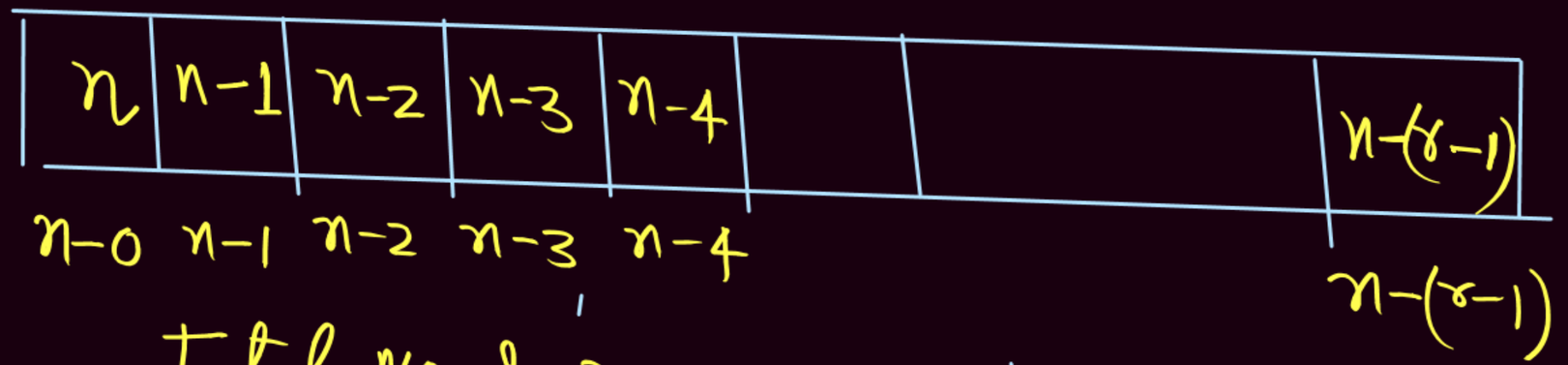
"n Diff. Items Taken r at a time"

n — r (Permutate) (Arrange)

W_1 job.

W_2 job.

$W = W_1 \times W_2$ they are simultaneously Together.



Total No. of ways $= n(n-1)(n-2) \cdots n-(r-1)$

$$= \frac{\lfloor n}{\lfloor n-r} = {}^{n}P_{r}$$

A	B	C	D
E	F		

6 Letters " Taken all at time = $6!$

" Taken 3 at a time $= {}^{n}P_r = {}^{6}P_3 = \frac{\lfloor 6}{\lfloor 6-3}$

6	5	4

$= 6 \times 5 \times 4$

$= 120$ ways.

$= \frac{\lfloor 6}{\lfloor 3} = \frac{6 \times 5 \times 4 \times \lfloor 3}{\lfloor 3}$

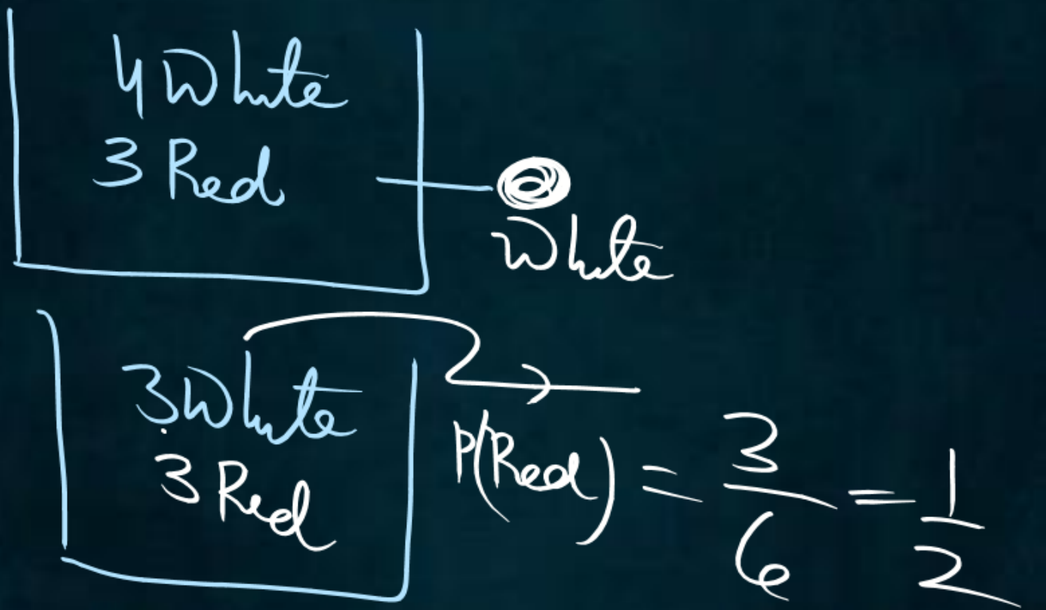
$= 120$

QUESTION

GATE PYQ

Time : 3 min

Q. A box contains 4 white balls and 3 red balls. In succession, two balls are randomly selected and removed from the box, given that the first removed ball is white, the probability that the second removed ball is red is_____.

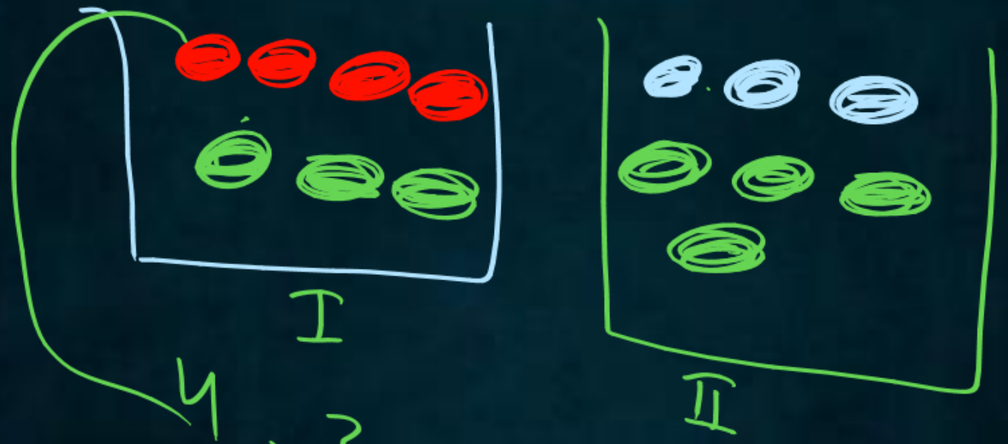


QUESTION

$\frac{12}{49}$

Time : 3 min

Q. There are two containers, with one containing 4 Red and 3 Green balls and the other containing 3 Blue balls and 4 Green balls. One ball is drawn at random from each container. The probability that one of the balls is Red and the other is Blue will be_____.



$\frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$

I st container Red X second contain blue

$= \frac{12}{49}$

QUESTION

Time : 3 min

Q. An urn contains 5 red and 7 green balls. A ball is drawn at random and its colour is noted. The ball is placed back into the urn along with another ball of the same colour. The probability of getting a red ball in the next draw is_____.

QUESTION

✓ IIT-advanced $= \frac{13}{32}$

Time : 3 min

Q. If from each of the three boxes containing 3 white and 1 black balls, 2 white and 2 black balls, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black ball will be drawn is_____.

3W 1B	2W 2B	1W 3B

WWB + BWW + WBW

$$\frac{3}{4}\times\frac{2}{4}\times\frac{3}{4} + \frac{1}{4}\times\frac{2}{4}\times\frac{1}{4} + \frac{3}{4}\times\frac{2}{4}\times\frac{1}{4}$$

$$= \frac{18+2+6}{64} = \frac{13}{32}$$ Ans

$W_1 W_2 W_3$ +

W W B + B W W + W B W

1st 2nd 3rd I II III I II III

= done.

QUESTION

Time : 3 min

Seven white balls and three black balls are randomly placed in a row. The probability that no two black balls are placed adjacently, equals:

A 1/2

B 7/15

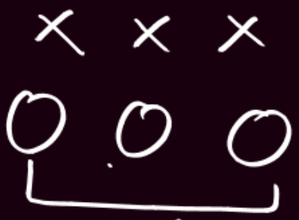
C 2/15

D 1/3

Total Position X =

favorable case $= {}^{8}C_{3} \times 3! \times 7!$

$$P(E) = \frac{{}^{8}C_{3} \times 3! \times 7!}{10!} \quad \text{Ans} = \frac{7}{15}$$



3 black

3 girls

NO 2 black balls ARE together

No Two girls Are sit together

Bar / cross method"

8 position 3 choose/ group

n DIFFERENT Items Selection at a time $= {}^{n}C_{r}$

9 girls 3 boys What is the prob. No two boys sit together

X | X | X | X | X | X | X | X | X | X | X

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩

$$fav = {}^{10}C_3 \times 3! \times 9!$$

$$P(E) = \frac{fav}{Total} = \frac{{}^{10}C_3 \times 3! \times 9!}{12!}$$

Bar | cross method

QUESTION

Time : 3 min

Three identical dice are rolled. The probability that the same number will appear on each of them, is:

(A) 1/6

(B) 1/36

(C) 1/18

(D) 3/28

(1,1,1)
(2,2,2)
(3,3,3)
(4,4,4)
(6,6,6)
(5,5,5)

fav. outcomes = 6

Die A + Die B + Die C

$$P(E) = \frac{6}{6 \times 6 \times 6} = \frac{1}{36}$$

QUESTION

Time : 3 min

$\frac{6}{6\times6\times6} = \frac{1}{36}$

Paragraph for Questions

Box I contains three cards bearing numbers 1, 2, 3 ; box II contains five cards bearing numbers 1, 2, 3, 4, 5; box III contains seven cards bearing numbers 1, 2, 3, 4, 5, 6, 7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the ith box i = 1,2,3.

Q. The probability that $x_1 + x_2 + x_3$ is odd, is:

(A) $\frac{29}{105}$

(B) $\frac{53}{105}$

(C) $\frac{57}{105}$

(D) $\frac{1}{2}$

1,2,3	1,2,3,4 5	1,2,3,4,5 6,7

x_i $i = 1, 2, 3$

$P(x_1 + x_2 + x_3 = \text{odd}) =$

$P(x_1+x_2+x_3 = \text{odd}) = \dfrac{\text{NUMBER of fav outcomes}}{\text{Total outcomes}}$

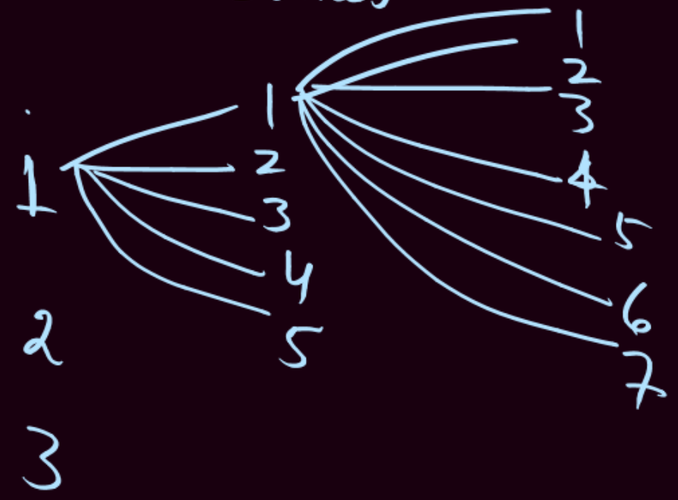
$W_1 \rightarrow 3$

$W_2 \rightarrow 5$

$W_3 \rightarrow 7$

$= 3 \times 5 \times 7$

$= 105 \text{ ways}$



because work Individual

I	II	III
1,2,3	1,2,3,4,5	1,2,3,4,5,6,7

Total No. of ways $= 3 \times 5 \times 7 = 105$

I II III

$O + O + O = odd \Rightarrow 2 \times 3 \times 4 = 24$

- ✓ EEO = 8
- ✓ OEE = 12
- ✓ EOE = 9

favourable outcomE = 53.

$$P(E) = \frac{53}{105}$$

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THANK YOU!

Here's to a cracking journey ahead!

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solutions + questions

SHEET

without ✓

with ✓